PORTFOLIO | TECHNICAL LEAD AND DATA ARCHITECT

Taha Abu Alata

I design and build data platforms on Google Cloud. I lead the team, and I write the code too.

Google Cloud Professional Cloud Architect | Google Cloud Generative AI Leader | Microsoft MCSE Cloud Platform | Google Project Management

BASED IN
Riyadh, Saudi Arabia

EXPERIENCE
15+ years

FOCUS
GCP, BigQuery, Customer 360



No confidential client data. All diagrams are my own. Dashboard numbers are sample data.

ABOUT ME

15 years of building platforms for Saudi organisations

15+ years in cloud, data and platforms

81 business applications documented

50+ government agencies on one platform

100 security controls mapped (PDPL, NCA, ISO)

5 to 15 engineers in the teams I led

1,500+ engineers trained in 20 countries

2010 to 2014
Money Experts
Finance training platforms, payments, pricing tools

2014 to 2019
Skillsoft and Oracle
Senior consultant and trainer, Oracle databases

2017 to 2018
Microsoft
Azure and Microsoft 365 cloud consultant

2020 to 2024
SAMI Advanced Electronics
Solution Architect and Team Lead

2024 to 2026
Qiddiya
Google Cloud operations, data and AI

2026 to now
AI contact centre
Principal Cloud and AI Architect

TODAY

Four projects, then my plan for your Customer 360

01

Qiddiya: Google Cloud data platform

BigQuery, Terraform, GKE, Vertex AI and executive dashboards

GCP · BigQuery · Looker Studio · Terraform

02

SAMI: national government platforms

One platform for 50+ agencies, shared data model, security

Architecture · APIs · Data model · Governance

03

Smart contact centre with AI agents

Voice and chat AI agents, data kept in Saudi Arabia, live KPIs

AI agents · RAG · Customer data · KPIs

04

Microsoft: many enterprise customers

Azure, identity, Power BI, Dynamics 365 and Power Platform

Azure · Power BI · Dynamics 365 · Security

05

Executive dashboards and pipelines

Dashboards from many sources, plus Dataflow, Airflow and dbt

Dataflow · Airflow · dbt · Checks

06

Customer 360: my plan

Design, matching customers, security, 13 weeks, the team

Golden record · PDPL · D365 · Power BI

1 The problem · 2 My role · 3 How it works · 4 The hard part · 5 The result · 6 What it means for you

PROJECT 01 | QIDDIYA INVESTMENT COMPANY | 2024 TO 2026

Google Cloud platform, data and AI for a giga-project

THE PROBLEM

Many cloud projects, teams and vendors. Leaders needed one clear answer: what is running, who owns it, what it costs, and is it secure?

MY ROLE

Assistant Manager, Cloud Operation. I owned the platform, the automation, the cost data and the internal AI tools. I designed them and built them myself.

WHAT I BUILT

- Terraform modules and CI/CD for GKE, Cloud SQL, network and IAM
- Shared VPC and secure network design
- Cost data model in **BigQuery** with **Looker Studio** dashboards
- A list of **81** applications, each with a design and an owner
- AI assistant with **Vertex AI, Gemini and RAG** on GKE
- Monitoring with Prometheus and Grafana

RESULTS

81	applications documented, each with a clear owner
1	cost view for leaders, checked against the bill
Live	AI assistant approved for production
IaC	every change through code and review

Hard part: each team deployed in its own way. **What I did:** shared Terraform modules, CI/CD and security checks. **For you:** the same standards from week 1 for a team of 6 to 8.

PROJECT 01 | HOW IT WORKS

From raw cloud data to decisions, in 4 steps

1 COLLECT

- Cloud billing export
- Asset list and labels
- Monitoring metrics
- Application list (81 apps)

2 LOAD

- Cloud Storage landing
- Scheduled loads and API pulls (Python)
- Check columns and row counts

3 MODEL IN BIGQUERY

- Raw: partitioned by date
- Clean: labels, owners, projects
- Business: cost per app, team, env
- Check: model total = bill total

4 USE

- Looker Studio executive dashboards
- Right-sizing and unusual spend reports
- Budgets and alerts

BUILD Terraform, CI/CD, code review

SECURE Least-privilege IAM, Shared VPC

AI Vertex AI, Gemini, RAG on GKE

Cost per app and team
every cost line has an owner

Right-sizing
found from usage vs size

Early alerts
no surprises at month end

Used by leaders
one dashboard for decisions

Hard part: labels were different in every project, so costs had no owner. **What I did:** fixed the labels in Terraform and joined the bill to the app list. **For you:** the same raw, clean, business layers, plus checks against the source.

PROJECT 01 | QIDDIYA UNIFIED DATA PLATFORM

How I built one data platform for the whole company, in 7 steps

1 Know the data
List every source, its owner and how fresh it must be

One owner per source

2 Land it raw
Copy as-is into Cloud Storage and BigQuery raw. Never edit.

Raw is the audit copy

3 Clean it
Fix types, names, dates and duplicates. Bad rows go aside.

Quality checks on load

4 Model it
Business tables with one definition for each number

Define a KPI only once

5 Protect it
Access by group, masked columns, audit logs, catalogue

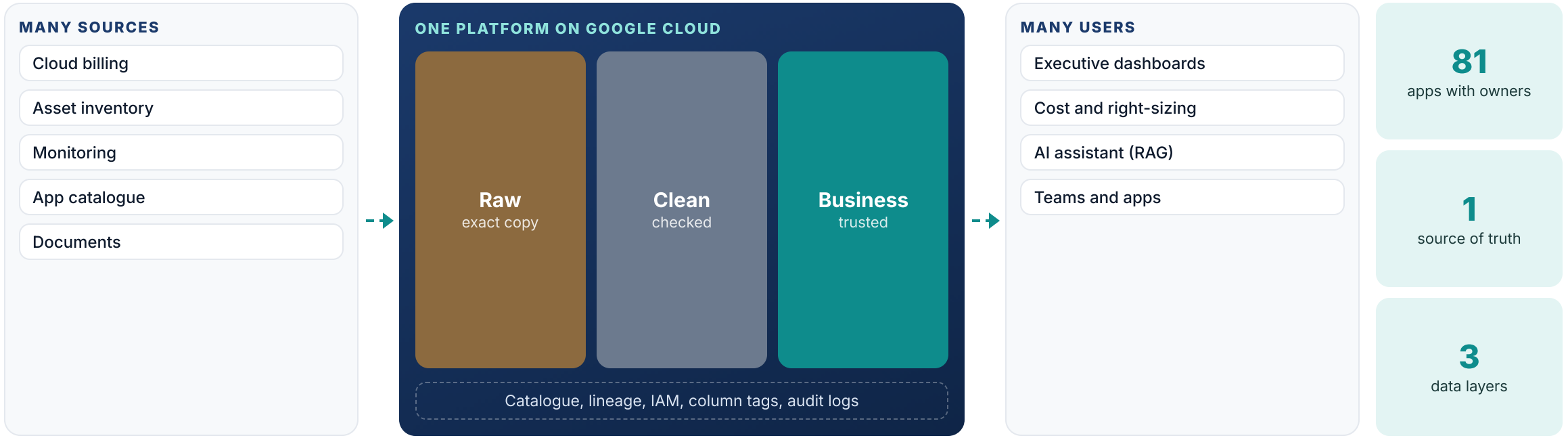
Least access by default

6 Serve it
Dashboards for leaders, APIs for apps, data for AI (RAG)

Same data for everyone

7 Run it
Terraform, CI/CD, monitoring, cost alerts, runbooks

Everything as code



81
apps with owners

1
source of truth

3
data layers

For you: Customer 360 follows the same 7 steps. The only new part is step 4: matching customers into one golden record.

PROJECT 02 | SAMI ADVANCED ELECTRONICS | 2020 TO 2024

National platforms: one shared model for many organisations

THE PROBLEM AND MY ROLE

I was Solution Architect and Team Lead for government and enterprise clients. This included the **DGA National Digital Transformation Platform** and four ministries. My teams had 5 to 15 engineers.

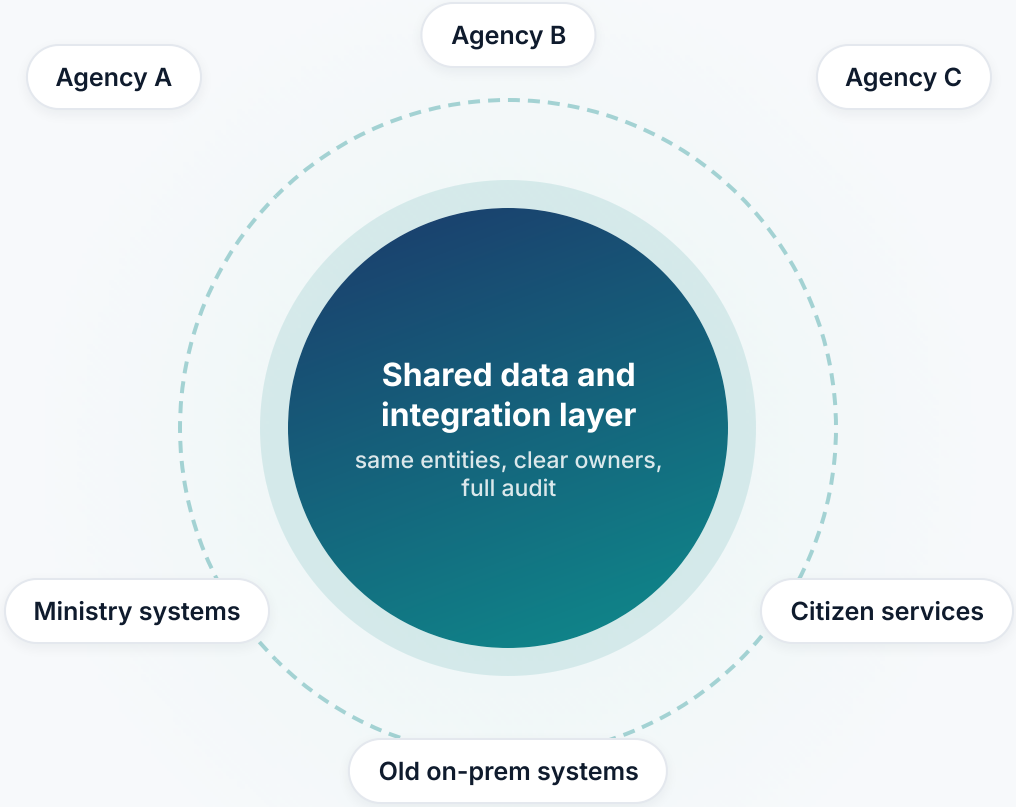
WHAT I DID, STEP BY STEP

1. Wrote the designs: HLD, LLD, decision records and interface documents
2. Designed the APIs (REST, OpenAPI) and event-based integration
3. Set performance, availability and recovery targets
4. Built security that meets NCA rules for sensitive data
5. Ran architecture review boards with all teams

50+
agencies onboarded

4
ministries

2022
Minister's recognition



For you: the hardest part of a golden record is the same. Everyone must agree on one customer model and on which system owns each field.

PROJECT 03 | SMART CONTACT CENTRE WITH AI AGENTS | 2026 TO NOW

Voice and chat AI agents for many customers, data kept in Saudi Arabia

1 CUSTOMER CONTACTS US

- Phone call
- Mobile app and web chat
- Outbound campaigns

2 AI AGENT ANSWERS

- Speech to text, AI agent, text to speech
- Agent uses tools (APIs) in a workflow
- Finds answers in the knowledge base (RAG)
- Guardrails block unsafe answers

3 CONNECT TO SYSTEMS

- Find the customer and their history
- Hand over to a human when needed
- Back-end APIs over a private, encrypted link

4 MEASURE

- Logs of every conversation and tool call
- KPI dashboards (CSAT, first-call fix)
- Cost per customer

MY ROLE

Principal Cloud and AI Infrastructure Architect. I own the cloud and AI design, the government links, cloud costs and Saudi compliance.

WHAT I BUILT

- Each customer fully separated, no keys in code
- Private VPN and API gateway with mutual TLS
- Moved the platform data into Saudi Arabia

FOR YOU

A contact centre agent must find the customer and see their history in seconds. Your Dynamics 365 agents need exactly this.

1 platform, 3 channels
voice, app and web

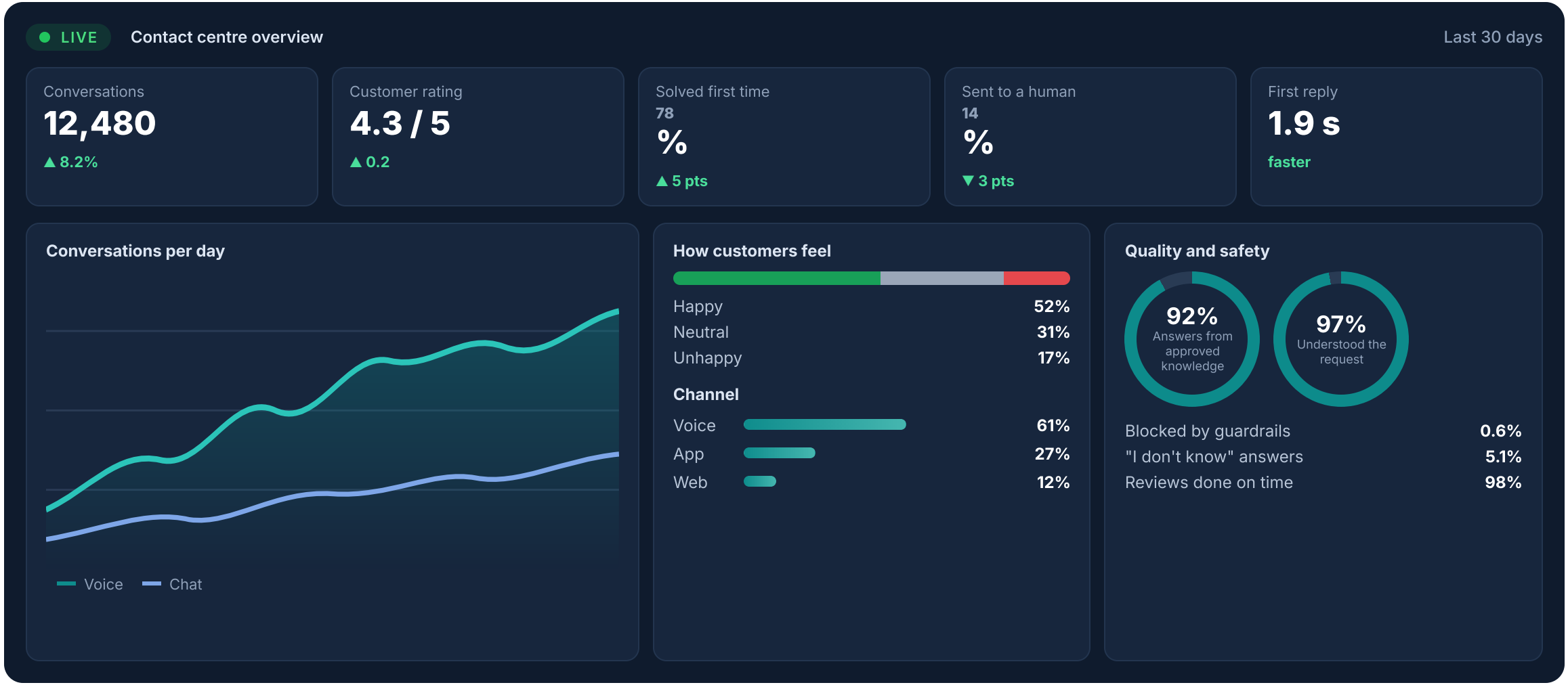
Data stays in KSA
move completed

Customers separated
own data and keys

Cost per customer
from real usage

PROJECT 03 | LIVE OPERATIONS DASHBOARD

What managers see: AI and human agents on one screen



Sample data made for this portfolio. It shows the dashboard design I build, not real client numbers.

PROJECT 04 | MICROSOFT | MANY ENTERPRISE CUSTOMERS

Microsoft cloud, identity, reports and CRM

THE PROBLEM AND MY ROLE

Cloud Architecture Consultant with Microsoft (2017 to 2018). I helped enterprise customers in different sectors, including financial services, move from their own servers to the cloud. I still work with Microsoft tools today.

WHAT I DELIVERED

- Azure designs, hybrid network and migration plans with costs
- Identity with Azure AD (Entra ID): single sign-on, MFA, Zero Trust
- Office 365, Exchange Online and SharePoint migrations
- Power BI reports with row-level security, embedded in apps
- Dynamics 365 Contact Center design, Power Apps and Power Automate

RECENT EXAMPLE: POWER PLATFORM PORTAL

Staff fill a Power Apps form. Data goes to a locked SharePoint list through a Power Automate flow. Staff never touch the data directly. Admins get dashboard views.

Identity
Entra ID, single sign-on, MFA, Conditional Access

Cloud
Azure IaaS and PaaS, hybrid network

Work tools
Microsoft 365, Exchange, SharePoint

Reports
Power BI, row and column security, embedded

CRM
Dynamics 365, Contact Center, Dataverse

Low-code
Power Apps, Power Automate

For you: I know both sides. We clean the data in BigQuery, then show it in **Power BI** and **Dynamics 365**. Each agent signs in with their own account, so they only see the rows they are allowed to see.

Dashboards I build that read from many sources

1 SOURCES

- Cloud billing APIs (multi-cloud)
- Platform usage APIs
- Resource tags and asset lists
- Contracts and credits
- HR, IT and Microsoft 365

2 PIPELINE

- Python collectors, read-only access
- Clean: currency, dates, owners
- Split shared costs by clear rules
- Check against the invoice. Stop if wrong.

3 MODEL

- BigQuery or workbook model
- Each measure defined once
- Every number can be traced

4 LEADERS

- Arabic and English dashboards
- Looker Studio, Power BI, SharePoint
- Monthly management pack

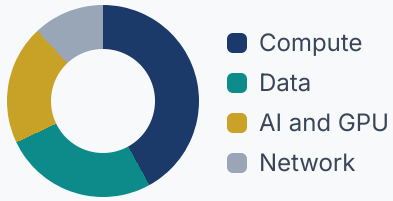
Monthly cloud spend

SAR 1.24M

✓ matches invoice

Cost by business unit

Unit A		38%
Unit B		26%
Unit C		18%
Shared		18%



Trust checks

- ✓ Model vs invoice under 0.5%
- ✓ Every cost has an owner
- ✓ Shows source and refresh time

Sample numbers. My rule: no number goes to a leader until it matches the source.

How I run Dataflow, Airflow, dbt and BigQuery in production

1 Land
Files land in Cloud Storage. Check columns and counts.

2 Dataflow
Clean, standardise, remove duplicates. Bad rows go aside.

3 BigQuery raw
Exact copy, never edited, partitioned by date.

4 dbt models
SQL models: staging, then business tables.

5 Quality gate
Tests and source checks. If they fail, stop and alert.

6 Publish
Refresh Power BI and send data to the CRM.

- Airflow (Cloud Composer) runs every step in order, retries failures, re-runs past days and sends alerts

DATAFLOW

- Batch and streaming
- Bad rows kept for review
- Monitor and re-process

AIRFLOW

- Safe to run twice
- Automatic retries
- Re-run any past date

DBT

- Load only new data
- Tests on every model
- Docs, lineage, CI on every change

BIGQUERY

- Partitions and clusters
- Access per dataset and column
- Cost limits per query

```
# Airflow DAG: nothing is published before checks pass
ingest >> dataflow_standardise >> dbt_run_staging
dbt_run_staging >> dbt_test_critical   # fail = stop + alert
dbt_test_critical >> match_identities >> dbt_run_marts
dbt_run_marts >> reconcile_vs_source >> [refresh_powerbi, sync_d365]
```

```
-- dbt: business table, loads only new rows
{{ config(materialized='incremental', unique_key='customer_sk',
          partition_by={'field': 'updated_date', 'data_type': 'date'},
          cluster_by=['customer_sk']) }}
select * from {{ ref('int_customer_golden') }}
{% if is_incremental() %}
where updated_date >= (select max(updated_date) from {{ this }})
{% endif %}
```

CUSTOMER 360 | MY PROPOSED DESIGN

Customer 360 on Google Cloud, with all data in Saudi Arabia

1 SOURCES

- Oracle data warehouse (OCI)
- GCP data lake
- Dynamics 365
- Policy and claims systems

2 BRING IN

- Datastream: only changes, private link
- Dataflow: batch or streaming
- Cloud Storage landing

3 BIGQUERY IN THE SAUDI REGION

- **Raw** — exact copy, never edited
- **Clean** — +966 mobiles, ID and Iqama, Hijri dates, Arabic names
- **Business** — golden record, policy, claim, interaction
- Locked zone for medical claims

4 USE

- Power BI with row security
- 360 view inside Dynamics 365
- BigQuery ML and Vertex AI

RUN Cloud Composer (Airflow)

TRANSFORM dbt with tests and source checks

SECURE IAM, own keys, column tags, audit logs

THE CUSTOMER MODEL

Customer	Policy	Claim	Provider
golden record	holder and insured	first notice to payment	hospital, garage

Interaction	Payment	Cross-reference
calls, cases, D365	premium, refund	golden ID to source IDs

First view based on the job description. I will confirm it with the source owners in the first weeks.

Finding the same customer in every system, in 5 steps

Step 1: Clean
mobile to +966, check ID and Iqama, name forms (Mohammed, Muhammad)

Step 2: Exact match
same National ID or Iqama, same CR number, policy + birth date

Step 3: Similarity score
name, birth date, mobile, email and city

Step 4: Decide
merge, send to a person, or keep apart

Step 5: Best value + links
keep every source ID, so we can trace and undo

EXAMPLE

CRM	Mohammed Al-Otaibi · 0551234567
POLICY	Muhammad Alotaibi · +966551234567
CLAIMS	محمد العتيبي · ID 10xxxxxxx

Score 0.96 → one golden record

0.92+	0.75 to 0.92	Below 0.75
Merge	Person checks	Keep apart

Example values and names. The business approves the real limits after testing.

My rule: the business approves the matching rules, not the engineers. Never merge two different people. In insurance that is a privacy incident.

CUSTOMER 360 | SECURITY FROM DAY ONE

PDPL, Saudi data residency and least access, in 6 layers

1
Location
Data can only be created in the Saudi region

2
Perimeter
VPC Service Controls, private links only

3
Encryption
Client-owned keys, TLS on every link

4
Access
Groups, one service account per pipeline, masked columns

5
Reports
Power BI row and column security

6
Proof
Audit logs that cannot be changed, with alerts

PDPL

Bring only the fields we need. Keep them only as long as allowed. When a customer asks for their data, the cross-reference table finds every record.

SENSITIVE DATA

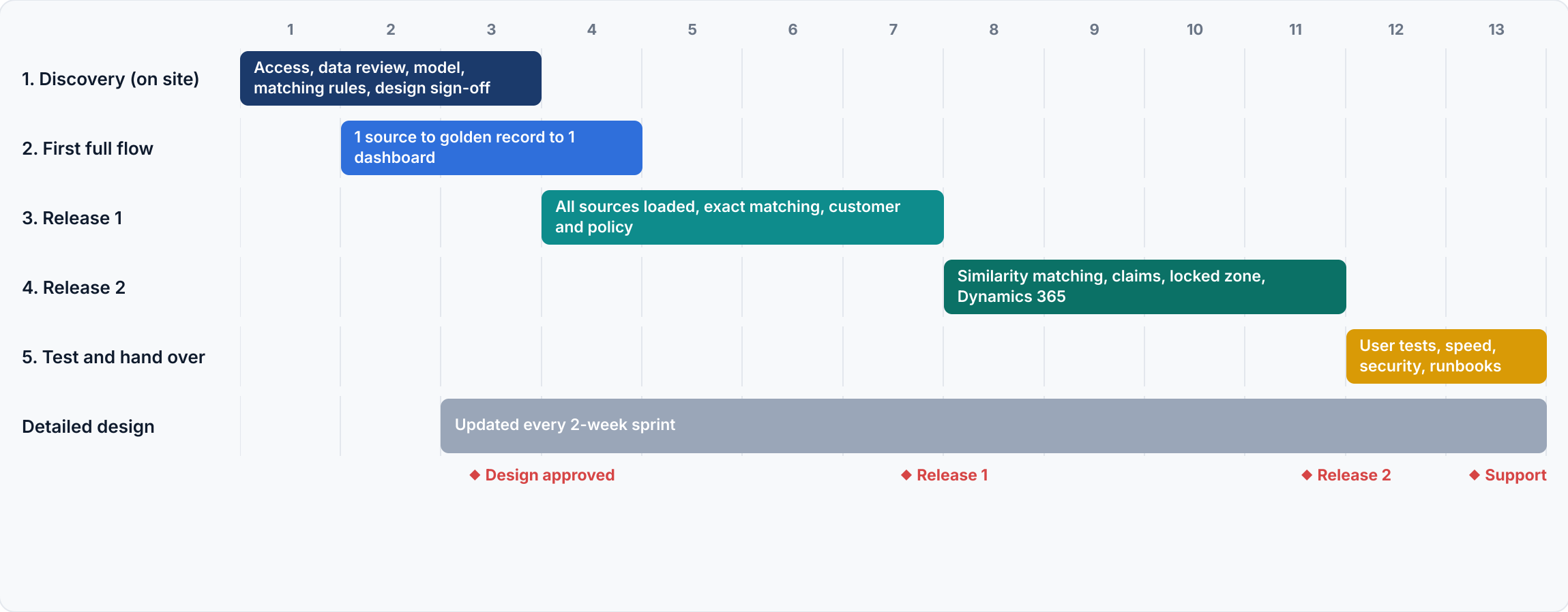
National ID, mobile, IBAN and medical data are tagged and masked. Medical claims sit in a separate locked project.

I HAVE DONE THIS

I mapped 100 controls to PDPL, NCA ECC, SDAIA, ISO 27001 and SOC 2 this year.

CUSTOMER 360 | DELIVERY PLAN

13 weeks: one full flow first, then two releases



MATCH RATE
% of customers with a golden record

DUPLICATES
how many before and after

MATCHES SOURCE
difference vs source system: target 0

USAGE
agents using the 360 view every day

Leading 6 to 8 engineers and still writing code

Technical Lead and Data Architect
design, hardest code, daily reviews

Data engineers (2 to 3)
loading, Dataflow, Airflow

Analytics engineers (2)
dbt, customer model, matching

Power BI developer
reports, row security

Dynamics 365 engineer
360 view, data sync

Tester (shared)
tests, user tests

Business owners for customer, policy and claims approve the rules

HOW I RUN THE TEAM

1. One owner for each area, 15-minute stand-up every day
2. Week 1: naming rules, dbt template, review checklist
3. No code goes live without a review and passing tests
4. I take the riskiest part myself, usually matching
5. One-page weekly update: done, next, risks, decisions
6. By week 13 the team can run it without me

"DONE" MEANS

- ✓ Code reviewed
- ✓ Matches source
- ✓ Access rules set
- ✓ dbt tests pass
- ✓ Docs updated
- ✓ Runbook written

EVIDENCE

Certificates you can check, and awards



Google Cloud Certified: Generative AI Leader
Issued Nov 2025, valid to Nov 2028



Google Agile Project Management
Check: coursera.org/verify/ZMFWYA505EGD

Google Cloud Professional Cloud Architect
Google Cloud

Google Project Management Professional Certificate
Check: coursera.org/verify/professional-cert/ASFNMBGD2RTA

Recognised by the Minister of Hajj and Umrah
Digital transformation of Hajj and Umrah services, 2022

Recognised by Princess Nourah University
AI and machine learning cloud platforms, 2023

Microsoft MCSE and MCSA
Cloud Platform and Office 365

THANK YOU

What I bring to your Customer 360

01

Speed

Discovery and design in the first 3 weeks. I have built this kind of platform many times.

02

Trust

Numbers that match the source. Security and PDPL built in from day one.

03

Handover

Clean code, tests and documents, so your team can run it after me.

04

Ready for AI

With a golden record in place, renewal prediction and call summaries come fast.

Questions?

engineer.taha.shaban@gmail.com | linkedin.com/in/taha-abu-alata